VDOVYDCHENKO ALEXANDER

email: keromvp@gmail.com telegram: @ker0sine
33 years old, MS, MBA. marital status: single. current location: Moscow, Russia

Profile

Hardware and software engineer with 10+ years of full-cycle development across embedded radioelectronics and telecommunications systems — from analog/RF and mixed-signal circuit design through FPGA/SoC firmware to embedded Linux BSP. Experience spans multichannel ADC/DAC data acquisition and processing systems, SoC/FPGA development, RF signal chain characterisation, compute and network platforms (OCP/19"). MBA in Strategic Management adds structured thinking at the intersection of engineering and product: architecture decisions, competitive benchmarking, build-vs-buy analysis, and cross-functional technical communication. Seeking a senior or lead role in an international or product-driven environment where deep technical ownership matters.

Objective: Seeking a senior or lead role in embedded systems, FPGA engineering, or system architecture within an international or product-driven environment — where deep technical ownership across the full HW-to-SW stack is valued. Open to complex embedded platforms, HPC, telecommunications, industrial or scientific instrumentation. Relocation and remote considered — looking for a team where engineering depth is respected and has direct impact on product decisions.

CORE SKILLS

HARDWARE	• Analog/Digital shematic design, PCB debug, ADC/DAC, SI/PI fundamentals, RF
SOFTWARE	• C, C++ (11–23), Rust, Python, POSIX, GDB, Valgrind, Perf, QEMU
EMBEDDED	• ARM, RISC-V, x86-64, BSP, Yocto, Buildroot, Petalinux, U-Boot, Linux kernel/modules
FPGA/SoC/HDL	• VHDL/(System)Verilog, Xilinx(AMD), RTL, Timing analysis, Gowin, Lattice, Anlogic
NETWORK	• Proxmox, ESXi, Linux & Windows network administration, virtualisation, OSI model
INSTRUMENTS	• VNA, spectrum & signal analysers, waveform generators, logic analyser, oscilloscope
BUSINESS	• Requirements analysis, competitive benchmarking, cust-dev, technical roadmap
LANGUAGES	• English — C1 Advanced (Cambridge EQF), German — B2 (Moscow State University) Russian — Native

BULAT / GAGARIN — OCP & 19" Compute Platforms ***Senior Embedded Engineer*** *2024 - present*

- ▷ System architecture analysis and design, competitive benchmarking, and technical solution selection;
- ▷ Platform bring-up and BSP development : Yocto build system, Linux kernel customisation, U-Boot, userspace daemons, REST API; both 19" (x86-64) and OCP form-factors.
- ▷ FPGA/CPLD design targeting server boards: RTL development in Verilog/VHDL for Anlogic, Gowin, and Lattice devices; functional simulation, synthesis, and hardware verification.
- ▷ Hardware bring-up: board-level debug, signal integrity checks, and functional verification of server platform HW/FW interactions.
- ▷ Test methodology authoring and acceptance, test documentation, product certification.

INSTRUMENTAL SYSTEMS — Radioelectronics ***Senior Embedded HW/SW Engineer*** *2020-2024*

- ▷ FPGA/SoC firmware on Xilinx (AMD) devices (Vivado, Vitis, ISE): DSP pipelines and filters, multichannel data acquisition, AXI and PCIe peripherals.
- ▷ Embedded Linux BSP for Zynq and standalone ARM/RISC-V targets: Buildroot, Yocto, Petalinux, Crosstool-ng;kernel module development, POSIX application layer (C/C++).
- ▷ Cross-compilation toolchain setup (GCC/Clang/MSVC, MSYS2/MinGW, QEMU); profiling and debugging with Perf, Valgrind, GDB.
- ▷ Customer requirements capture, technical specification authoring, test methodology development; cross-department coordination, task decomposition, staff mentoring.

INSTRUMENTAL SYSTEMS — Radioelectronics ***Engineer — Analog / RF / FPGA*** *2016-2020*

- ▷ Analog and RF schematic design: multichannel ADC/DAC front-ends, HF signal conditioning, impedance matching; simulation.
- ▷ Design SI analysis — SNR, SFDR, Phase Noise, Jitter, Eye Diagram; characterisation with VNA, spectrum/signal analysers, Lock-In.
- ▷ FPGA firmware modules (Verilog/VHDL) for signal processing on Xilinx; lab prototype validation.
- ▷ Component selection and BOM management across Analog Devices, Texas Instruments, Xilinx, Maxim, Linear Technology, NXP, Marki Microwave etc.

INSTITUTE OF RADIO-ENGINEERING AND ELECTRONICS — Radiophysics ***Research Fellow*** *2017-2019*

- ▷ Scientific research on superconducting THz receivers and SSPDs (unique samples); low-temperature physics experiments, statistical data processing, and technical report preparation.

INSTRUMENTAL SYSTEMS ***Junior Engineer / Test Engineer*** *2014-2016*

- ▷ Testing and acceptance of RF/HF modules, ADC/DAC boards, system and application soft and firmware, radar sub-systems and antenna units: functional, thermal, and vibration testing;
- ▷ Production and FAT methodology authoring.

FORMAL EDUCATION

○ Moscow International Higher Business School — MIRBIS

oct 2023 - aug 2025
(2 years)

Master of Business Administration • Strategical Management

Strategic Management, Market Analysis & Research, Strategic Development

• Market analysis • Strategic Development • Business Architecture • Management

○ Moscow Institute of Physics and Technology (MIPT)

sep 2017 - jul 2019
(2 years)

Master of Science • Solid State Electronics and Information Systems

Department of Solid State Electronics and Information Systems • Applied Mathematics & Physics

• Scientific Research • Statistical Processing • Radio-Physics & Electronics

○ Moscow Technical University of Communications and Informatics (MTUCI)

sep 2009 - aug 2014
(5 years)

Specialist Degree • Radio-Electronics

Department of Radio Receiving and Transmitting Devices • Radio-Electronics

• Embedded systems • CPU/FPGA/DSP • Analog & Digital HF/RF • Schematic Design

LANGUAGES

English — C1 Advanced. Cambridge English Qualifications (CEFR C1)
German — B2. Moscow State University · 2 years · Professional & cultural
Russian — Native

PROFESSIONAL DEVELOPMENT & CERTIFICATES

System & Low-Level Programming

▷ Modern Linux Kernel Programming (6 months) — 2025-2026
Kernel modules · Linux internals · C

▷ Modern C Programming (6 months) — 2022
x86-64 · ARM · Low-level programming · POSIX · Make/CMake

▷ Advanced Linux-Based System Administration (7 months) — 2021-2022
**nux Administration · Network · Virtualisation · Automatization · Security*

Software Engineering

▷ C++ Professional (6 months) — 2023-2024
C++ 11–23 · STL · Boost · Mocking/Testing · Cross-platform · Networking

▷ Rust developer (6 months) — 2022-2023
Network · Concurrent & Parallel · Async · Functional programming

▷ Program Architecture & Design Patterns (6 months) — 2023
Software Architecture · Microservices · Design Patterns · OOP

▷ Python Programming (6 months) — 2021
Web · SQL/NoSQL · CI/CD · Git · Django/Flask · REST

Infrastructure & Domain-Specific

▷ Data Centre Fundamentals — 2025
High-load infrastructure · Data centre architecture

▷ MATLAB & Simulink for Processes and System Modelling — 2020
MATLAB · Simulink · System modelling

▷ Measurements & Test Methods Creation — 2018
Measurement methodology · Test design · Metrology